

ZHUOWEI XU

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Skills

Programming: C++, Python, MATLAB/Simulink, Julia

Hardware / Fabrication: CAD Design, PCB Design, 3D Printing (Bambu Lab, Raise3D, Formlabs), Arduino

Robotics / ML: Robot Operating System (ROS), MuJoCo, Imitation Learning, Tactile Sensing, Robot Manipulation

Education

Carnegie Mellon University

Sept. 2024 – May 2026

Master of Science in Mechanical Engineering, GPA: 3.9/4.0

Pittsburgh, PA

Coursework: Mechanics of Manipulation, Robot Dynamics, Modern Controls, Computer Vision, Machine Learning

Shanghai Jiao Tong University

Aug. 2020 – Jul. 2024

Bachelor of Mechanical Engineering

Shanghai, China

Professional Experience

Fraunhofer Project Center for Smart Manufacturing at SJTU

Jun. 2023 – Sept. 2023

Research Intern

Shanghai, China

- Developed an air-chamber-actuated underactuated finger that lifted 200 g at 0.1 MPa.
- Improved actuation performance via exoskeleton stiffness enhancement and variable-stiffness joint design.

Shanghai BOCHU Electronic Technology

Nov. 2023 – May 2024

Research Intern

Shanghai, China

- Built a galvo mirror-robot co-work system, extending planar galvo workspace to 3D using a 6-DoF robotic arm.
- Built a kinematic calibration model with error compensation, identifying system parameters from planar images.

Projects

Tactile-Augmented Imitation Learning Policy for Fine-Grained Manipulation

May 2025 – Mar. 2026

Carnegie Mellon University, Robotics Institute

- Built a **multimodal imitation learning framework** that fuses vision, force, and vibrotactile sensing.
- Developed a **kinematic-twin-based teleoperation system** for whole-body control of the robotic hand and Franka arm, with a MuJoCo virtual twin for state monitoring and control command visualization.
- Trained and deployed policies for automated grasping of fragile and contact-sensitive objects, where force-aware policies achieved a **50% higher success rate** on the potato chip pinching task than the vision baseline.

Learning Tactile Event Extraction for Object-Centric Manipulation

Sept. 2025 – Apr. 2026

Carnegie Mellon University, Robotics Institute

- Developed a **top- k tactile event extraction** module that selects informative key frames from long-horizon exploration histories, matching a Transformer-based history compressor with a more compact representation.
- Extracted **localized object-centric tactile features** from vibrotactile exploration, enabling property-aware decisions, selecting the smooth sponge side for wiping with **100% success** and **100% transfer** to unseen sponge.
- Extended event-based tactile reasoning to **multi-object manipulation**, achieving **100% accuracy** in selecting target objects with specified properties and outperforming policies without tactile context by **75 %**.

Multi-modal Tactile Fingertip Design for Robotic Hands

Oct. 2024 – Sept. 2025

Carnegie Mellon University, Robotics Institute

- Developed a low-cost multimodal tactile fingertip integrating 90 Hz strain-gauge force sensing and 44.1 kHz vibrotactile sensing, running reliably on lightweight hardware over three months of experiments.
- Enabled **adaptive grasping within 0.3 N of the minimum required grasping force** and reconstructed localized tactile features for contact-rich manipulation.
- Achieved **98% accuracy** in fully occluded object recognition through multimodal fusion and achieved a **60% higher manipulation success rate** than a YOLO-based vision baseline.

Model-Based Trajectory Control and Planning for a Whiteboard Robot

Feb. 2025 – May 2025

Course Project, Optimal Control and Reinforcement Learning, Carnegie Mellon University

- Built real-time dynamics and control for a whiteboard-erasing robot, with a visual tracking system for target localization.
- Implemented a 2D simulation environment and benchmarked PID, MPC, and TVLQR under noise; validated on hardware, with MPC achieving the best tracking performance.

Publications

- **Zhuowei Xu**, Zilin Si, Kevin Lee Zhang, Oliver Kroemer, Zeynep Temel. “A Multi-modal Tactile Fingertip Design for Robotic Hands to Enhance Dexterous Manipulation.” IEEE/RSJ International Conference on Intelligent Robots. arxiv.org/pdf/2510.05382.
- **Zhuowei Xu**, Zilin Si, Kevin Lee Zhang, Moonyoung Lee, Shashwat Singh, Oliver Kroemer, Zeynep Temel. “A Low-Cost Tactile Fingertip Design for Dexterous Robotic Hands.” ICRA 2025 Workshop “Handy Moves: Dexterity in Multi-Fingered Hands.” openreview.net/forum?id=5kj8uifi2w.
- Yongzhou Long, Zhuang Zhang, **Zhuowei Xu**, Enlin Gu, Qiuji Lu, Hao Wang, Genliang Chen. “Lightweight and Powerful Vacuum-Driven Gripper with Bioinspired Elastic Spine.” IEEE Robotics and Automation Letters. ieeexplore.ieee.org/document/10287554.